Midterm Project 2– Part 2

Task 3(a)

The table on the left below exemplifies performance data obtained for a water heat pipe like that analyzed in Task 1. These are data points for operation indicating $T_{f,cond\ in}$, $\dot{q}$ and the resulting T_{chip} value. These are all for the following operating constant values listed to the right below:

$T_{f,cond\ in}$ (degC)	$\dot{q}$ (W)	T_{chip} (degC)
8.0,	50.0	14.3
8.0,	100.0	19.6
8.0,	200.0	32.4
8.0,	250.0	40.2
10.0	50.0	17.1
10.0	150.0	28.3
10.0	200.0	34.7
10.0	250.0	39.3
10.0	300.0	47.7
10.0	400.0	59.4
12.0	100.0	24.3
12.0	200.0	37.1
12.0	400.0	60.9
14.0	100.0	27.0
14.0	250.0	43.4
14.0	400.0	62.3
16.0	100.0	27.1
16.0	150.0	34.2
16.0,	300.0	52.2

Parameters of interest

(i) Set or compute heat pipe design and evaporator structure parameters:

D_p = 0.015 #m

L_{evap} = 0.025 #m

L_{adiab} = 0.30 #m

L_{cond} = 0.025 #m

$L_{eff} = L_{adiab} + 0.5\ L_{evap} + 0.5\ L_{cond}$

t_w = 0.0025 #m (multiple layers - mesh)
#wick: 100 mesh copper screen

$A_w = (\pi/4)[D_p^2 - (D_p - 2t_w)^2]$ #wick cross-sectional area

$(UA)_e$ = 12.7 #W/K evaporator conductance

$D_v = D_p - 2t_w$

$A_v = (\pi/4)(D_p - 2t_w)^2$ # m^2

$r_{p,\min}$ = 0.000063 #m minimum pore radius

κ = 2.3 x 10^{-10} # m^2 permeability

θ = 10. # interface contact angle, degrees

$r_{e,\min} = r_{p,\min} / \cos(\theta)$ #minimum interface radius achievable

(ii) Set condenser design and operating parameters

$T_{f,cond\ in}$ = 15.5 +273. #K – condenser coolant in

$\dot{q}$ = 40.0 #W -heat transfer rate

$\dot{m}_c$ =0.0080 # kg/s

ε_c = 0.72 condenser heat exchanger effectiveness ($\dot{q} = \dot{m}_c c_{p,f} \varepsilon_c (T_2 - T_{f,cond\ in})$)

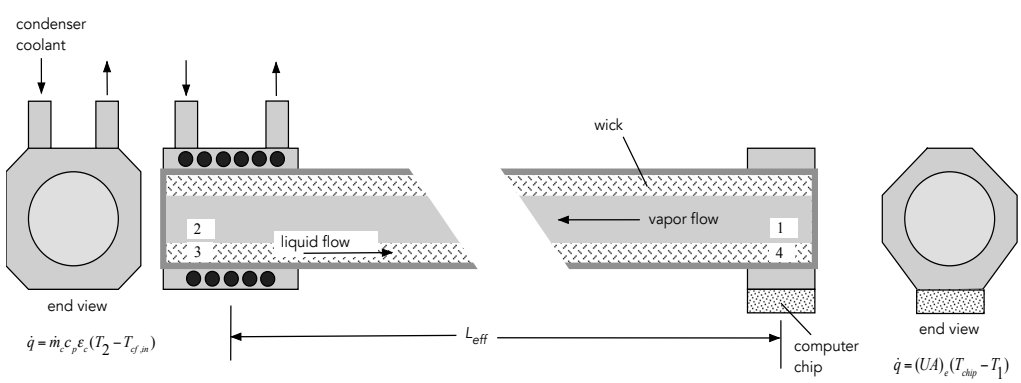
$c_{p,f}$ = 4180 J/kgK

Make a copy of the code used in Task 1 to use as a starting point for this task. Modify this code so that values of $(UA)_e$ and ε_c can be specified together with $T_{f,cond\ in}$, $\dot{q}$ and the resulting T_{chip} value from one of the rows of the table. The code should be modified to output the absolute value of the difference between the data chip temperature and the model predicted chip temperature $|T_{chip,data} - T_{chip,pred}|$.

Task 3 (b) When the code derived from the program developed in Task 1 is assembled, run the code and predict $|T_{chip,data} - T_{chip,pred}|$ for at least 36 combinations of $(UA)_e$ and ε_c spanning the following ranges:

$$5.7 < (UA)_e < 17.7$$
$$0.2 < \varepsilon_c < 0.95$$

Randomly pick data in a row of the table above to submit with the chosen $(UA)_e$ and ε_c values. Use the results to construct a table or surface plot of $|T_{chip,data} - T_{chip,pred}|$ vs. $(UA)_e$ and ε_c spanning the above ranges. From the table or plot, locate the region in the $(UA)_e$ and ε_c space where the $|T_{chip,data} - T_{chip,pred}|$ is smallest. Make some additional calculations in that region and determine your best estimates for the $(UA)_e$ and ε_c values that minimize the $|T_{chip,data} - T_{chip,pred}|$. List all of your computed $(UA)_e$, ε_c, $|T_{chip,data} - T_{chip,pred}|$ combinations in a table. Finally, pick 5 random rows of data in the table and determine how small $|T_{chip,data} - T_{chip,pred}|$ is for these combinations of the heat transfer rate and condenser water in temperature for the estimated best $(UA)_e$ and ε_c values. Summarize the results in a table.



Task 4 (a)
A proposal is made to use methanol for the heat pipe design described in Part 1. Make a copy of the program you developed in Part 1 and make changes in that program to convert it to the

program for this Task. You will be converting the program to model a heat pipe similar to that in Task 1, but with some design changes so that it can use methanol as a working fluid. At the end of this write-up you will find a table of saturated liquid and vapor properties for methanol.

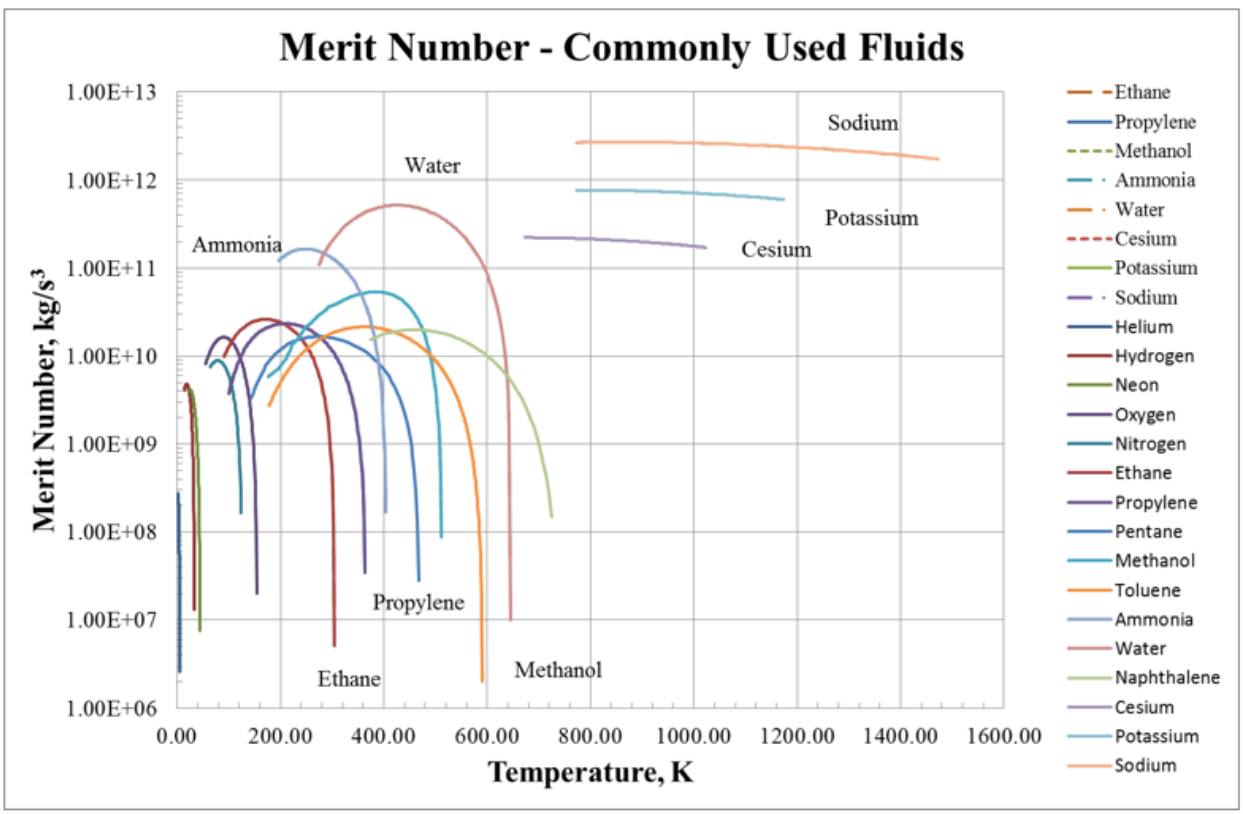
Why use methanol? There are two reasons it may be an attractive option. One is that, unlike water, it does not freeze until it reaches -97.6 °C. The second is that it has a favorable merit number in temperature ranges corresponding to applications of interest.

Heat pipes fluids are ranked by the Merit Number formula:

$$N_l = \frac{\rho_l \sigma \lambda}{\mu_l}$$

where:

ϱ_l	Liquid density
σ	Surface tension
λ	Latent heat
μ_l	Liquid viscosity



From: https://www.1-act.com/thermal-solutions/passive/heat-pipes/heat-pipes-101

In the copy of your program for this task, modify the program to model the heat pipe with methanol, and set the heat pipe design and operating parameters by doing the following:

(i) Set or compute heat pipe design and evaporator structure parameters:

D_p = 0.015 #m

L_{evap} = 0.025 #m

L_{adiab} = 0.30 #m

L_{cond} = 0.025 #m

L_{eff} = L_{adiab} + 0.5 L_{evap} + 0.5 L_{cond}

t_w = 0.0025 #m (multiple layers - mesh) #wick: 100 mesh copper screen

$A_w = (\pi/4)[D_p^2 - (D_p - 2t_w)^2]$ #wick cross-sectional area

$(UA)_e$ = 10.8 #W/K evaporator conductance

$D_v = D_p - 2t_w$

$A_v = (\pi/4)(D_p - 2t_w)^2$ # m^2

$r_{p,\min}$ = 0.000045 #m minimum pore radius

κ = 3.0 x 10^{-10} # m^2 permeability

θ = 10. # interface contact angle, degrees

$r_{e,\min} = r_{p,\min} / \cos(\theta)$ #minimum interface radius achievable

(ii) Set condenser design and operating parameters

$T_{f,cond\ in}$ = 15.5 +273. #K

$\dot{q}$ = 80.0 #W

$\dot{m}_c$=0.0080 # kg/s

ε_c= 0.6 condenser heat exchanger effectiveness ($\dot{q} = \dot{m}_c c_{p,f} \varepsilon_c (T_2 - T_{f,cond\ in})$)

$c_{p,f}$= 4180 J/kgK

(iii) Change the working fluid property calculations to compute properties for methanol

Replace all the water property values at the 15, 30, 45 and 60 degree C temperatures with the corresponding methanol values at 10, 30, 50 and 70 degrees C and replace the water saturation pressure values at these temperatures with the corresponding values for methanol at those temperatures:

$P_2 = P_{sat}(T_2) = \exp\{C_0 - C_1 / T_2\}$, where for methanol, use saturation pressure values for methanol at 10°C and 50 °C to get C_0 and C_1 values for methanol.

And replace water values with methanol values in:

```
#PSEUDO COMPUTER CODE:
PA=0.10*100,000.  #Pa   Psat at 10 deg C
PB=0.25*100,000.  #Pa   Psat at 30 deg C
PC=0.55*100,000.  #Pa   Psat at 50 deg C
PD=1.31*100,000.  #Pa   Psat at 70 deg C

#below property10 is the saturated value at 10 °C, likewise for 30, 50, 70
rhov10 =
rhov30 =
rhov50 =
rhov70 =
   ⋮
etc., and:

WA = (P2-PB)*(P2-PC)*(P2-PD)/( (PA-PB)*(PA-PC)*(PA-PD) )
WB = (P2-PA)*(P2-PC)*(P2-PD)/( (PB-PA)*(PB-PC)*(PB-PD) )
WC = (P2-PA)*(P2-PB)*(P2-PD)/( (PC-PA)*(PC-PB)*(PC-PD) )
WD = (P2-PA)*(P2-PB)*(P2-PC)/( (PD-PA)*(PD-PB)*(PD-PC) )
rhov = WA*rhov10 + WB*rhov30 + WC*rhov50 + WD*rhov70
rhol = WA*rhol10 + WB*rhol30 + WC*rhol50 + WD*rhol70
muv = WA*muv10 + WB*muv30 + WC*muv50 + WD*muv70
mul = WA*mul10 + WB*mul30 + WC*mul50 + WD*mul70
sig = WA*sig10 + WB*sig30 + WC*sig50+ WD*sig70
hlv = WA*hlv10 + WB*hlv30 + WC*hlv50 + WD*hlv70
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Task 4 (b) Once the methanol model is assembled, run the code for methanol as a working fluid with the values of the design parameters and operating condition parameters set in Task 4a and determine whether the heat pipe design specified in Task 4a above can reach a stable operating point at a heat transfer rate of 80 W using methanol as a working fluid. If it can, summarize the operating parameters and state point conditions for steady operation in a table, and assess the impact of using methanol on the heat transfer rate under the design conditions. If a stable operation is not achieved, try heat flux levels between 80 W and 30 W to see if a stable operating point can be achieved at heat flux values in that range. Describe what you did in your report, and present a summary of results in a table to justify your conclusions about what heat transfer rate levels are achievable for this design and conditions.

Task 4 (c) Try running your code for the following heat rejection rates (in W) from the chip $\dot{q}$: 20, 30, 50, 60. For each value where a stable point is achieved, determine the chip temperature T_{chip} in °C and the effective evaporator interface r_e in μm. Plot both T_{chip} and r_e versus chip heat transfer rate

Task 4 (d) For the operating conditions in 4(a), if you adjust the minimum pore radius and permeability of the wick, can you achieve a 30% increase in heat transfer rate? Report in a table

the $r_{p,min}$, κ, and the $\dot{q}$ achieved and the $r_{e,min}$ and r_e values obtained for your modified design . Note that you must have $r_e > r_{e,min}$ for a design that works.

Thermophysical properties of methanol

Methanol (methyl alcohol) is a chemical compound with chemical formula CH_3OH.
Thermophysical properties for temperatures ranging -50-150 °C are indicated in the table below.

Temperature (°C)	Latent Heat (kJ/kg)	Liquid Density (kg/m³)	Vapor Density (kg/m³)	Liquid Thermal Conductivity (W/m°C)	Liquid Viscosity ($x10^3$ N·s/m²)[1]	Vapor Viscosity ($x10^5$ N·s/m²)[2]	Vapor Pressure (bar)[3]	Vapor Specific Heat (kJ/kg°C)	Liquid Surface Tension ($x10^2$ N/m)[4]
-50	1194	844	0.01	0.210	1.700	0.72	0.01	1.20	3.26
-30	1187	834	0.01	0.208	1.300	0.78	0.02	1.27	2.95
-10	1182	819	0.04	0.206	0.945	0.85	0.04	1.34	2.63
10	1175	801	0.12	0.204	0.701	0.91	0.10	1.40	2.36
30	1155	782	0.31	0.203	0.521	0.98	0.25	1.47	2.18
50	1125	764	0.77	0.202	0.399	1.04	0.55	1.54	2.01
70	1085	746	1.47	0.201	0.314	1.11	1.31	1.61	1.85
90	1035	724	3.01	0.199	0.259	1.19	2.69	1.79	1.66
110	980	704	5.64	0.197	0.211	1.26	4.98	1.92	1.46
130	920	685	9.81	0.195	0.166	1.31	7.86	1.92	1.25
150	850	653	15.9	0.193	0.138	1.38	8.94	1.92	1.04

[1] multiply number in table by 10^{-3} to get liquid viscosity in N·s/m²
[2] multiply number in table by 10^{-5} to get vapor viscosity in N·s/m²
[3] multiply number in table by 100 to get vapor pressure in kPa
[4] multiply number in table by 10^{-2} to get surface tension in N/m